Literal presentations and local periods in hexagonal extensions

An application to Fernandes–Leemans–Ivić Weiss

The retained primary is the 21-page author edition of *Hexagonal extensions of toroidal maps and hypermaps* [1]. We preserve its parameter order, literal presentations and regular/chiral conventions. The original constructions and questions are attributed to its authors. All group orders below are full *type-preserving* automorphism orders; correlations are separate.

The complete earlier marked-kernel common paper and the complete new *Rotation certificates and affine period families* supplement are paid premises. The latter contains the entire accepted cubic, square and exact finite proofs formerly located in the L121 application, now shared with this application. No sibling application proof is free. The standard torus and faithful affine Coxeter conventions remain explicit classical inputs. We do not reprove the prior classifications quoted as source Tables 1–2 or Theorems 3.1–3.3.

1 One certificate for each literal finite row

For each parameter row let $P = \langle x, y, z \mid \mathcal{R} \rangle$ be exactly the source's rotation presentation. The entire six presentation builders occur in the required program `presentation_engine.g`. Parameters are read from `table_roster.json`, not reconstructed from a claimed order. Put $a_0 = 1, a_1 = x, a_2 = y, a_3 = z$ and $P_i = \langle a_j^{-1} a_k : j, k \neq i \rangle$. The engine establishes an exact isomorphism from the abstract presentation to its permutation group. This is an upper-bound computation as well as a finite image. In T8.8, complete coset enumeration over $\langle y, z \rangle$ gives index 17280; the abstract local presentation has order 150. The resulting permutation image has order 2592000, equal to $17280 \cdot 150$, so the upper and lower bounds coincide. ACE is bounded by 500000 cosets,5000000 workspace words and 60 seconds. The first general attempt was stopped at 120 seconds; its receipt is not treated as evidence against finiteness.

For every row the engine checks all 256 intersections of the 16 pair-difference subgroups, all 16 normalized flag counts, generation for each of the four rank-three residues and the six actual rank-two residue incidence graphs. The shared theorem gives thinness, residual connectivity and precisely two rotation-group chamber orbits. If simultaneous inversion satisfies every literal relator it is an involutory automorphism. The faithful two-copy semidirect completion is checked for all 256 marked intersections and 16 coset-Helly identities. Otherwise a specified failed inverse relator certifies chirality and that P itself is the full type-preserving group.

Theorem 1 (The computed literal rows)**.** *All 66 finite rows below define finite regular or chiral hypertopes, with the exact full orders and actual marked local groups displayed. Each has the source's asserted regular/chiral status. The full orders in source Table 5 rows 8 and 10 must be doubled for the literal presentations in this retained edition.*

Proof. Apply the shared rotation and local-kernel propositions to the exact per-row outputs. The abstract upper bound and generator maps are separate from the subgroup and flag tests. Each local group is identified by an onto map from its complete two-generator presentation with matching order; its full translation kernel is also computed. The regular completion has order $2|P|$ and the chiral full group has order $|P|$. No assertion is transferred between rows merely because their numerical orders agree. The required software and entire row roster are explicit computational premises. □

R and C mean regular and chiral. The four local orders are rotation orders in omitted-type order 0,1,2,3. Seven inherited-family rows are handled separately. Printed structural descriptions are audited separately below. They are never identified merely from an order.

Row	Literal parameters	Printed g	Proved g	Type	Local rotation orders
T3.1	1, 2; 1, 2	1,344	1,344	C	12/42/12/42
T3.2	1, 2; 2, 1	20,160	20,160	C	12/42/12/42
T3.3	2, 0; 1, 2	672	672	C	12/42/12/24
T3.4	2, 0; 1, 3	4,368	4,368	C	12/78/12/24
T3.5	2, 0; 1, 4	16,128	16,128	C	12/126/12/24

Row	Literal parameters	Printed g	Proved g	Type	Local rotation orders
T3.6	$2,0;\ 2,3$	13,680	13,680	C	12/114/12/24
T4.1	$2,0;\ 2,0$	240	240	R	6/24/6/24
T4.2	$2,0;\ 6,0$	720	720	R	6/72/6/24
T4.3	$3,0;\ 3,0$	2,916	2,916	R	6/54/6/54
T4.4	$3,0;\ 4,0$	241,920	241,920	R	6/96/6/54
T4.5	$3,0;\ 1,1$	324	324	R	6/18/6/54
T4.6	$3,0;\ 2,2$	41,472	41,472	R	6/72/6/54
T4.7	$1,1;\ 1,1$	108	108	R	6/18/6/18
T4.8	$2,2;\ 2,2$	13,271,040	13,271,040	R	6/72/6/72
T4.9	$3,0;\ 1,3$	33,696	33,696	C	6/78/6/54
T4.10	$4,0;\ 1,2$	12,096	12,096	C	6/42/6/96
T4.11	$6,0;\ 1,2$	756,000	756,000	C	6/42/6/216
T4.12	$1,2;\ 3,6$	2,016	2,016	C	6/126/6/42
T4.13	$1,2;\ 4,4$	36,288	36,288	C	6/288/6/42
T4.14	$1,2;\ 6,6$	2,268,000	2,268,000	C	6/648/6/42
T4.15	$3,5;\ 2,1$	672	672	C	6/42/6/42
T4.16	$1,4;\ 2,1$	2,016	2,016	C	6/42/6/126
T4.17	$1,2;\ 1,2$	672	672	C	6/42/6/42
T5.1	$2,0;\ 2,0;\ 2,0$	240	240	R	6/24/24/12
T5.2	$2,0;\ 3,0;\ 3,0$	1,296	1,296	R	6/54/54/12
T5.3	$2,0;\ 4,0;\ 4,0$	15,360	15,360	R	6/96/96/12
T5.4	$2,0;\ 6,0;\ 2,2$	2,880	2,880	R	6/72/72/12
T5.5	$3,0;\ 2,0;\ 2,0$	1,296	1,296	R	6/24/24/27
T5.6	$3,0;\ 2,0;\ 3,0$	13,824	13,824	R	6/24/54/27
T5.7	$3,0;\ 2,0;\ 4,0$	165,888	165,888	R	6/24/96/27
T5.8	$3,0;\ 2,0;\ 5,0$	2,592,000	5,184,000	R	6/24/150/27
T5.9	$3,0;\ 2,0;\ 2,2$	3,888	3,888	R	6/24/72/27
T5.10	$3,0;\ 2,0;\ 4,4$	248,832	497,664	R	6/24/288/27
T5.11	$3,0;\ 3,0;\ 1,1$	972	972	R	6/54/18/27
T5.12	$3,0;\ 1,1;\ 1,1$	324	324	R	6/18/18/27
T5.13	$3,0;\ 1,1;\ 3,3$	2,916	2,916	R	6/18/162/27
T5.14	$3,0;\ 2,2;\ 2,0$	3,888	3,888	R	6/72/24/27
T5.15	$4,0;\ 2,0;\ 2,0$	15,360	15,360	R	6/24/24/48
T5.16	$2,0;\ 1,2;\ 2,1$	336	336	C	6/42/42/12
T5.17	$2,0;\ 1,3;\ 2,5$	2,184	2,184	C	6/78/78/12
T5.18	$2,0;\ 1,4;\ 1,2$	336	336	C	6/42/42/12
T5.19	$2,0;\ 1,4;\ 3,6$	8,064	8,064	C	6/126/126/12
T5.20	$2,0;\ 2,3;\ 3,2$	6,840	6,840	C	6/114/114/12
T5.21	$3,0;\ 1,2;\ 1,2$	275,562	275,562	C	6/42/42/27
T6.1	$2,0;\ 2,0$	384	384	R	4/24/24/12
T6.6	$3,0;\ 1,1$	144	144	R	4/18/18/12
T6.7	$6,0;\ 1,1$	576	576	R	4/72/18/12
T6.8	$3,0;\ 1,3$	58,968	58,968	C	4/54/78/12
T6.9	$1,2;\ 1,2$	2,688	2,688	C	4/42/42/12
T6.10	$1,2;\ 2,1$	1,008	1,008	C	4/42/42/12
T6.11	$1,2;\ 3,1$	58,968	58,968	C	4/42/78/12
T7.1	$3;\ 3,0;\ 1,1$	360	360	R	12/12/18/18
T7.2	$3;\ 6,0;\ 1,1$	23,040	23,040	R	12/12/18/72
T7.3	$3;\ 1,2;\ 1,1$	1,512	1,512	C	12/12/18/42
T7.4	$3;\ 1,4;\ 1,1$	90,720	90,720	C	12/12/18/126
T7.5	$3;\ 2,4;\ 1,1$	774,144	774,144	C	12/12/18/168
T7.6	$4;\ 2,0;\ 1,1$	4,320	4,320	R	24/24/18/24
T7.7	$4;\ 1,1;\ 1,1$	3,456	3,456	R	24/24/18/18
T8.4	$2,0;\ 1,1;\ 4,0;\ 3,0$	165,888	165,888	R	96/54/18/24
T8.5	$3,0;\ 2,0;\ 1,1;\ 2,0$	1,296	1,296	R	18/24/24/54
T8.6	$3,0;\ 2,0;\ 1,1;\ 3,0$	13,824	13,824	R	18/54/24/54
T8.7	$3,0;\ 2,0;\ 1,1;\ 4,0$	165,888	165,888	R	18/96/24/54
T8.8	$3,0;\ 2,0;\ 1,1;\ 5,0$	5,184,000	5,184,000	R	18/150/24/54
T8.9	$2,0;\ 1,2;\ 1,1;\ 6,0$	15,120	15,120	C	18/72/42/24
T8.10	$2,0;\ 1,2;\ 1,2;\ 2,0$	352,800	352,800	C	42/24/42/24
T8.11	$2,0;\ 1,2;\ 1,1;\ 0,3$	7,620,480	7,620,480	C	18/504/42/24

2 Actual local periods and corrections

The standard torus presentations have orders $6(a^2+ab+b^2)$, or $3(a^2+ab+b^2)$ for the triangular hypermap. Twelve ordinary period relations become stronger local quotients in the global presentation. For instance, T7.1 has actual local rotation order 18 where $(3,0)$ would give 54. Each exact local kernel is recorded below, in the ordered translation words of the shared local-kernel proposition. It specifies the marked quotient, not an unlabelled torus selected by order.

Row	Omitted type	Nominal local order	Actual order	Exact kernel basis
T4.2	1	216	72	$(6,0);(2,2)$
T4.12	1	378	126	$(21,0);(4,1)$
T4.15	3	294	42	$(7,0);(2,1)$
T5.4	1	216	72	$(6,0);(2,2)$
T5.17	2	234	78	$(13,0);(3,1)$
T5.18	1	126	42	$(7,0);(2,1)$
T5.19	2	378	126	$(21,0);(4,1)$
T6.6	1	54	18	$(3,0);(1,1)$
T6.7	1	216	72	$(6,0);(2,2)$
T7.1	3	54	18	$(3,0);(1,1)$
T7.2	3	216	72	$(6,0);(2,2)$
T8.9	1	216	72	$(6,0);(2,2)$
T8.11	1	not a period	504	$(42,0);(8,2)$

There is an additional literal issue in source Section 8, physical p.17. Its final h-word is $y^{-1}(yz)^2 = zyz$, whose cube is already trivial from $y^3 = z^2 = 1$. For T8.11, $v = (0,3)$ thus adds no period. The literal global group nevertheless has the printed order 7620480. Its actual local rotation order is 504, with kernel basis $(42,0),(8,2)$ and determinant 84. We certify this actual quotient from the three triangle relations and its full lattice kernel, without inserting a likely intended different word. All other positive Table 8 rows have $h = 0$.

T5.8 has rotation order 2592000 and regular full order 5184000; T5.10 has rotation order 248832 and regular full order 497664. Both pass every intersection, flag and actual-residue test. The printed full totals are 2592000 and 248832. Their structural abbreviations cannot be asserted unchanged when their implied orders conflict; the exact marked presentations and local models specify the corrected objects.

Adding the actual local kernel relations changes no global group, since they already hold in it. Conversely the original period relations hold in each exact local quotient. Thus the same global group is universal for these *actual* local presentations. A claim with uncollapsed preassigned periods is stronger and is not made for the listed exceptions.

3 Shared affine families and inherited finite rows

For Table 6 rows 2–5 and all $s \geq 3$, the complete shared cubic proof gives E/sD of full order $48s^3$, its marked residues and a faithful degree-$4s$ CPR representation. In its notation (a,b,d,c) use

$$x = ba, \qquad y = bc, \qquad z = bd.$$

Then $(y^{-1}z^{-2})^2 = (c(bd)^2b)^2$ is the fixed local relation, and $(x^{-1}z^{-2})^s = (a(bd)^2b)^s$ is the variable relation. The ordered parameter pair is $((2,0),(s,0))$. The full cubic toroid has order $96s^3$; the common proof retains its explicit index-two embedding and the source CPR labels. The endpoint $s = 2$ is separately certified as T6.1, not inferred from the faithful-$4s$ argument.

Table 8 row 1 is the entire shared square family with parameters $((2,0),(1,1),(s,s),(2,0))$ and full order $1152s^4$, for $s \geq 1$. The fixed affine group, integral parabolic lattices, period kernel, intersections and flag transitivity are proved in the paid common proof. Table 8 rows 2–3 are the complete shared finite cases of orders 720 and 46080. Their actual first local period has the degree-three quotient proved there, rather than the uncollapsed $(3t,0)$ type. The shared orientation-adapter proof gives exact rotation/reflection presentation identities and explicit marker permutations alongside the unchanged proof migration; matching orders alone is not the argument.

4 Structural columns and corrected extension claims

Every one of the 73 printed group descriptions is retained in the ledger. V denotes one of the 29 previously verified named isomorphisms. F denotes one of the 35 complete recursive normal-subgroup, quotient and

complement certificates in the whole structural companion [5]. S denotes one of the seven shared affine or finite structural proofs in [4]; O records a printed order incompatible with the literal universal group. The two complete companion proofs and their whole required programs and input data are part of this proof boundary. An unnamed action does not remove the obligation to prove normality, the quotient, or a stated splitting. Conversely, verifying a bracketed extension shape does not claim it uniquely identifies all abstract groups with those factors.

Thirty-four F rows verify the printed shape under explicit brackets. For T4.8, the asserted split stage must instead be dotted:

$$\big((2^{5+6}.(A_5 \times C_3)) : S_3\big) : S_3.$$

The companion supplies both the corrected positive shape and the obstruction to the printed splitting. Its perfect terminal derived subgroup of order 120 cannot have an index-two subgroup isomorphic to A_5; the complete subgroup argument is retained there. The seven S rows have explicit normal factors and complements, including the uniform square-family splitting for every positive parameter. No new geometry is inferred solely from these structural descriptions.

The required `structural_groups.py` constructs the same full group from the certified rotation generators and the checked inversion automorphism. For the 29 V rows it constructs the stated target and verifies an actual bijective group homomorphism. It records the source generators, their images and the target generators. The target denoted $A \times_{C_2} S_3$ is the subgroup of $A \times S_3$ where the two quotient signs agree, using $A/A' = C_2$; this is the "fibre" notation below. The group $\mathrm{P\Sigma L}_2(27)$ is the preimage of the order-three subgroup of $\mathrm{Aut}(\mathrm{PSL}_2(27))/\mathrm{PSL}_2(27)$. For $\mathrm{P\Sigma L}_2(49)$ the program explicitly adjoins the field map $u \mapsto u^7$ to the fractional-linear $\mathrm{PSL}_2(49)$ action. This distinguishes the field action from the inequivalent diagonal extension $\mathrm{PGL}_2(49)$. Wreath products use their full stated permutation action. Classical finite-group constructors and exact isomorphism algorithms are declared software premises; no source extension action is inferred from composition factors.

For T8.11 a smaller factor certificate replaces an expensive direct isomorphism search. The two minimal normal subgroups have orders 2520 and 504. Their opposite centralizers have orders 5040 and 1512, commute, intersect trivially, and generate the full group. Separate bijections identify them with S_7 and $\mathrm{Aut}(\mathrm{PSL}_2(8))$. Multiplication therefore proves the displayed direct-product identification. The first bounded direct search timed out; no failure claim was inferred.

Row	Printed group column	Exact scope here
T3.1	$2 \,.\, L_2(7) : 2 : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T3.2	$2 \,.\, A_7 : 2 : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T3.3	$L_2(7) : 2 \times 2$	V: PGL(2,7) x C2
T3.4	$L_2(13) \times 2 \times 2$	V: PSL(2,13) x C2 x C2
T3.5	$Q_8 \,.\, (L_2(7) \times 2) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T3.6	$L_2(19) : 2 \times 2$	V: PGL(2,19) x C2
T4.1	$A_5 : 2 \times 2$	V: S5 x C2
T4.2	$S_5 \times S_3$	V: S5 x S3
T4.3	$3^{1+2} \times 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.4	$\mathrm{Aut}(L_3(4))$	V: Aut(PSL(3,4))
T4.5	$3^{1+2} : 2 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.6	$2^{1+6} : 3^2 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.7	$3^{1+2} : 2^2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.8	$2^{5+6} : A_5 \times 3 : S_3 : S_3$	F: corrected nonsplit stage; complete companion proof.
T4.9	$L_3(3) : S_3$	V: Aut(PSL(3,3)) fibre_C2 S3
T4.10	$U3(3) : 2$	V: Aut(PSU(3,3))
T4.11	$U3(5) : S_3$	V: Aut(PSU(3,5))
T4.12	$2 \,.\, L_2(7) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.13	$U3(3) : S_3$	V: Aut(PSU(3,3)) fibre_C2 S3
T4.14	$3 \,.\, U3(5) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.

Row	Printed group column	Exact scope here
T4.15	$2 \,.\, L_2(7) : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.16	$2 \,.\, L_2(7) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T4.17	$2 \,.\, L_2(7) : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.1	$S_5 \times 2$	V: S5 x C2
T5.2	$S_3 \times S_3 \times S_3 : S_3$	V: S3 wreath S3
T5.3	$2^{1+6} : (A_5 : 2)$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.4	$S_5 \times 2^2 : S_3$	V: S5 x S4
T5.5	$6 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.6	$2^{1+6} : 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.7	$2^{1+6} : 6 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.8	$2 \,.\, (A_5 \times (A_5 \times A_5)) : 3 : 2$	O: printed order incompatible; no structural identification asserted.
T5.9	$S_3 \times 3 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.10	$2^{1+6} : (3 : S_3 : S_3 : 3) : S_3$	O: printed order incompatible; no structural identification asserted.
T5.11	$3^{1+2} : 6 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.12	$3^2 \times S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.13	$3^{1+2} \times 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.14	$S_3 \times 3 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.15	$4 \,.\, (2^4 : A_5) : 2 \times 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.16	$L_2(7) : 2$	V: PGL(2,7)
T5.17	$L_2(13) \times 2$	V: PSL(2,13) x C2
T5.18	$L_2(7) : 2$	V: PGL(2,7)
T5.19	$Q_8 \,.\, L_2(7) : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T5.20	$L_2(19) : 2$	V: PGL(2,19)
T5.21	$3^{2+6} : 7 : 3 : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T6.1	$2^{1+4} \times 2 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T6.2	$S_3 \times S_3 \times S_3 : S_3$	S: complete shared normal-factor and complement proof.
T6.3	$2^{1+6} : 2^2 : S_3$	S: complete shared normal-factor and complement proof.
T6.4	$5^3 : 2 : 2 : 2 : S_3$	S: complete shared normal-factor and complement proof.
T6.5	$2^3 : S_3 : S_3 : S_3 : S_3$	S: complete shared normal-factor and complement proof.
T6.6	$S_3 \times 2^2 : S_3$	V: S3 x S4
T6.7	$2^4 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T6.8	$L_2(27) : 3 : 2$	V: Aut(PSL(2,27))
T6.9	$2^6 : 7 : 3 : 2$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T6.10	$L_2(7) \times 3 : 2$	V: PGL(2,7) x C3
T6.11	$L_2(27) : 3 \times 2$	V: PSigmaL(2,27) x C2
T7.1	$A_5 : S_3$	V: S5 fibre_C2 S3
T7.2	$2^4 : A_5 : 2 : 2 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T7.3	$L_2(8) : 3$	V: Aut(PSL(2,8))

Row	Printed group column	Exact scope here
T7.4	$L_2(8) \times A_5 : 3$	V: Aut(PSL(2,8)) x A5
T7.5	$2^9 \,.\, L_2(8) : 3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T7.6	$A_6 : 2 \times S_3$	V: S6 x S3
T7.7	$2^{1+4} : 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.1	$s^4 : 2^{1+4} : S_3 : S_3$	S: complete shared normal-factor and complement proof.
T8.2	$A_5 \times 2 : S_3$	S: complete shared normal-factor and complement proof.
T8.3	$2^4 : A_5 : 2 \times 2 : 2 : S_3$	S: complete shared normal-factor and complement proof.
T8.4	$2^{1+6} : S_3 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.5	$S_3 \times S_3 \times S_3 : S_3$	V: S3 wreath S3
T8.6	$2^{1+6} : 3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.7	$2^{1+6} : S_3 : S_3 : S_3 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.8	$2 \,.\, (A_5 \times (A_5 \times A_5)) : 2 : S_3$	F: full normal-factor and splitting certificate; explicit brackets in companion.
T8.9	$3 \times S_7$	V: C3 x S7
T8.10	$L_2(49) \times 3 : 2$	V: PSigmaL(2,49) x C3
T8.11	$S_7 \times (L_2(8) : 3)$	V: S7 x Aut(PSL(2,8))

5 The diagram list and its limit

The ten patterns in source Figure 1 represent sixteen fixed edge-labelled cases after expanding its parameter $p \in \{3, 4, 5, 6\}$. Here is the complete elementary reduction. A connected rank-four nonlinear diagram has one of five underlying graphs: a star, a triangle with a pendant edge, a four-cycle, K_4 minus one edge, or K_4. Every rank-three triangle must be spherical or Euclidean, and at least one must be Euclidean. The retained triangle criterion is $1/p + 1/q + 1/r \geq 1$, with equality for a toroidal face and omitted edges assigned label two. An edge with label greater than six would have a neighbouring edge of label at least three by connectedness, violating this criterion. Thus labels two through six suffice.

A star must have arm labels $(3, 3, 6)$, $(3, 4, 4)$ or $(4, 4, 4)$. A triangle with a pendant edge has all triangle labels three and pendant label three through six. In a four-cycle, a label six forces its neighbours to be three and permits the opposite label three through six. With no six, an affine adjacent pair must be $(4, 4)$; labels five cannot then occur. The possibilities are an adjacent pair of fours and two threes, three fours and one three, or four fours. Finally, in K_4 minus an edge and in K_4, all edges belong to triangles and hence all have label three. This gives respectively $3 + 4 + 7 + 1 + 1 = 16$ cases, exactly the source's ten parametrized patterns. The optional rational enumeration in `diagram_types.py` checks this case split; it is not a premise of the written argument. The reduction is a list of possible diagrams, not a proof that every case has a finite universal hypertope.

6 The isolated nonorientable example

The complete separate note [6] verifies the source's pp. 18–19 assertion about $\{\{4, 4\}_{(4,0)}, \{4, 6\}_3\}$. The literal universal group has order 768 and local orders 128 and 48. All 256 marked subgroup intersections and the local universal maps are checked. The vertex figure has face counts $(4, 12, 6)$ and a closed flag walk of odd length nine, so it has nonorientable genus four. The entire two-page argument and its exact finite-presentation program are required parts of this application; the displayed permutation image alone is not used as a universal upper bound. This is verification of an isolated source assertion, not an answer to a subsequent open question.

7 Full source boundaries and questions

Section 2's coset construction and C-plus terminology retain Tits and the earlier-literature attributions. The shared theorem supplies the finite rotation-to-geometry direction with its actual hypotheses. Source Tables

1–2 and Theorems 3.1–3.3 quote earlier classifications and remain retained inputs. Diagram (4)'s asserted presentation reduction is distinguished from diagram (5)'s explicitly experimental existence claim. The complete correlation companion [7] proves inverse presentation maps and both parameter transformations for every positive integer parameter. For each existing regular universal fork hypertope, its actual prescribed residues give the corresponding universal string polytope and recover the fork marking. Its correct rank-four generator list contains the fourth generator omitted in the source parenthesis. For diagram (5), the exact group and conditional intersection-property claims are proved, while the experimental flag-transitivity and existence claim remains qualified. There is no unrestricted converse from a parity character alone. The whole four-page companion is part of this application. Bounded failed searches in Section 8 are documentary, not nonexistence theorems. The source's principal finite tables concern the first three diagrams.

Problems 9.1–9.2 and Conjectures 9.1–9.3 are separate from positive-row recovery. Previously reviewed Problem 9.2 work constructs finite quotient geometries for every integer parameter and proves the literal universal group infinite for all parameters at least six. Its exact old proofs and reviews are retained separately; no fresh solution, priority or current-open status is claimed. The four small universal orders now have the T8.5–T8.8 certificates. Finite quotients do not imply finite universal groups. None of these computations proves the source lists exhaustive or proves its negative conjectures.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, retained 21-page author edition (2018 institutional file).

[2] *Rotation certificates and affine period families*, complete supplement to *Supported kernels, translation lattices and finite marked quotients*, private proof migration.

[3] M. E. Fernandes, D. Leemans, C. A. Piedade and A. Ivić Weiss, *Two families of locally toroidal regular 4-hypertopes arising from toroids*, twelve-page author edition.

[4] *The seven shared group structures in the hexagonal tables*, complete accepted shared structural companion, four pages.

[5] *Checking the structural descriptions of finite hexagonal extensions*, complete accepted structural companion, four pages.

[6] *A locally toroidal polytope with a nonorientable vertex figure*, complete accepted companion, two pages.

[7] *Correlation reductions and the exact square-torus parameters*, complete accepted companion, four pages.

Checking the structural descriptions of finite hexagonal extensions

Human Compression project

Abstract

This supplement addresses structural group assertions in the retained source on hexagonal extensions. Exact marked presentations and incidence computations do not by themselves identify an asserted extension. We use their already certified permutation images to exhibit normal subgroups, identify quotients and check the indicated splittings. No presentation enumeration or geometric census is repeated here. Thirty-four printed shapes are verified under explicit brackets. One split-extension assertion is disproved and replaced by a certified nonsplit shape.

1 What must be proved

The source explicitly adopts the group notation of the Atlas (physical page 3). A string containing a colon asserts a split extension; a product sign asserts a direct product. Those assertions remain mathematical obligations even when the action is unnamed. In the certificates below an iterated expression is fully parenthesized. Unless parentheses in the source dictate otherwise, extensions are read from the left; every direct product is explicitly bracketed. We prove the resulting shape and do not claim it uniquely determines the group among all abstract groups of that shape.

The ambient group is not chosen by its order. We use the saved marked rotation group of the same literal source row. For a chiral row that is the full type-preserving group. For a regular row we use the previously checked inversion automorphism α and its faithful action on two copies: g acts as $(g, \alpha(g))$ and the new involution exchanges the copies. Its order is twice the rotation-group order. The earlier universal-presentation and geometry certificates establish that this is the required group; they are retained whole as premises of this supplement.

Lemma 1 (Structural witnesses)**.** *Let G be a finite group. Suppose $N \lhd G$, with N of a certified shape A and G/N of a certified shape B. Then G has extension shape $A.B$. If moreover a subgroup $C \leq G$ satisfies*

$$|N \cap C| = 1, \qquad |N||C| = |G|,$$

then $G = N \rtimes C$ and $C \cong G/N$. If also $[N, C] = 1$, then $G = N \times C$. These tests can be applied recursively to all the parenthesized factors.

Proof. The quotient homomorphism has kernel N and image G/N. Its restriction to C is injective. The order identity makes it surjective and gives $NC = G$. Thus conjugation by C supplies the action in the semidirect product. In the commuting case multiplication $N \times C \to G$ is a bijective homomorphism. Recursion applies these facts to the indicated subgroups and quotient groups, rather than inferring the factors from orders. □

Named simple and symmetric factors are checked by actual bijective homomorphisms to the declared standard groups. An elementary factor p^a is checked to be elementary abelian of order p^a. A factor p^{a+b} used here is certified with

$$\mathrm{Z}(P) = [P, P] = \mathrm{Frat}(P), \quad |\mathrm{Z}(P)| = p^a, \quad |P/\mathrm{Z}(P)| = p^b,$$

with both central subgroup and quotient elementary abelian. When $a = 1$ this proves that P is extraspecial. The exponent and actual centre, derived and Frattini subgroups are retained in the witness, so no stronger choice of extraspecial type is inferred from an unsigned source symbol. For Q_8 we check noncommutativity, order eight and exactly one involution; the two nonabelian groups of order eight are then distinguished directly. Cyclic factors are checked cyclic, not merely assigned by their order.

2 Finite implementation and trust

The required program reconstructs the previously accepted group and searches normal subgroups of each prescribed order. A candidate is accepted only after the entire kernel and quotient shapes are checked. For a colon it then obtains an actual complement and independently checks the two order identities in Lemma 1. For a direct product it also checks commutation on generators. The program retains the subgroup and complement generators, not just a printed abstract description.

The normal-subgroup search can first pass to a smaller quotient. If the required quotient has derived length at most d, its kernel contains $G^{(d)}$. If its solvable radical is trivial, the kernel contains the solvable radical of G. A normal p-subgroup lies in $O_p(G)$, so a p-power kernel can instead be found among the subgroups of that core which are normal in G. These are necessary conditions, not order-based identifications; all the asserted witness properties are checked afterward.

For index two, a complement is an involution outside the kernel. It is enough to examine a Sylow 2-subgroup, because conjugation preserves membership in a normal subgroup. For an S_3 quotient, a complement has a subgroup of order three. Conjugate this into a Sylow 3-subgroup and consider its conjugacy-class representatives there. For each representative y outside the kernel, an involution x inverting y must lie in the normalizer of $\langle y \rangle$. Conjugate it into a Sylow 2-subgroup of that normalizer. That conjugation sends y to y or y^{-1} and preserves the required relation. Thus these finite searches cover every possible complement. A direct-product complement lies in $C_G(N)$; it is a complement of $Z(N)$ there, provided $NC_G(N) = G$.

Normal-subgroup, quotient, Frattini, complement and homomorphism computations use GAP's exact finite-group algorithms. This is a declared computational premise, not a sample argument. A failed or timed-out search is not counted as a structural disproof. The certificate applies only to its exact row and literal presentation. The seven parameter-dependent or shared L121 rows are being treated in a separate supplement, and are not inferred from these finite rows.

3 The correction in Table 4, row 8

The source prints $2^{5+6} : A_5 \times 3 : S_3 : S_3$ for the row with both parameters $(2, 2)$, of full order 13 271 040. Its typography was checked directly on physical page 12. Every expression built from these factors using only split extensions and direct products contains a subgroup isomorphic to its A_5 factor, regardless of the ambiguous association. The certified group does not contain such a subgroup.

Indeed its derived series has orders

$$13\,271\,040, \quad 3\,317\,760, \quad 368\,640, \quad 122\,880, \quad 1\,920, \quad 120.$$

The final subgroup P satisfies $P' = P$. Every perfect subgroup K of G lies in every term of the derived series: from $K \leq G^{(j)}$ we obtain $K = K' \leq G^{(j+1)}$. Hence any A_5 subgroup would lie in P and have index two there. An index-two subgroup is normal and gives a nontrivial abelian quotient of P, contradicting $P' = P$. As a second exact obstruction, the complete

enumeration of the 120 elements of P gives exactly one involution; A_5 has fifteen. The recorded order histogram is

$$(1:1),\ (2:1),\ (3:20),\ (4:30),\ (5:24),\ (6:20),\ (10:24).$$

This is a finite certificate for the particular exact group, not a failure of a search for complements.

The positive corrected shape, with all association specified, is

$$\big((2^{5+6}.(A_5 \times C_3)) : S_3\big) : S_3. \tag{1}$$

The program exhibits a normal special subgroup of order 2^{11}, identifies the indicated quotient $A_5 \times C_3$, and gives actual complements at both outer colons. The inner dotted extension is nonsplit, since a complement would again contain A_5. The order, exact marked presentation and geometry of this source row are unchanged. We do not replace the printed expression silently or claim a new historical result.

4 Complete finite-row ledger

The table gives the exact bracketed assertions certified here. A dot records an extension; a colon records the actual complement checks. Only T4.8 is a replacement of the source's split claim. The p^{a+b} notation has the explicit special-group meaning stated above.

Row	Certified shape
T3.1	$(((C_2 \,.\, \mathrm{PSL}(2,7)) : C_2) : C_2)$
T3.2	$(((C_2 \,.\, A_7) : C_2) : C_2)$
T3.5	$((Q_8 \,.\, (\mathrm{PSL}(2,7) \times C_2)) : S_3)$
T4.3	$(((3^{1+2} \times C_3) : S_3) : S_3)$
T4.5	$((3^{1+2} : C_2) : S_3)$
T4.6	$(((2^{1+6} : 3^2) : S_3) : S_3)$
T4.7	$(3^{1+2} : 2^2)$
T4.8	$(((2^{5+6} \,.\, (A_5 \times C_3)) : S_3) : S_3)$ (corrected)
T4.12	$((C_2 \,.\, \mathrm{PSL}(2,7)) : S_3)$
T4.14	$((C_3 \,.\, \mathrm{PSU}(3,5)) : S_3)$
T4.15	$((C_2 \,.\, \mathrm{PSL}(2,7)) : C_2)$
T4.16	$((C_2 \,.\, \mathrm{PSL}(2,7)) : S_3)$
T4.17	$((C_2 \,.\, \mathrm{PSL}(2,7)) : C_2)$
T5.3	$(2^{1+6} : (A_5 : C_2))$
T5.5	$(((C_6 : S_3) : S_3) : S_3)$
T5.6	$(((2^{1+6} : C_3) : S_3) : S_3)$
T5.7	$((((2^{1+6} : C_6) : S_3) : S_3) : S_3)$
T5.9	$(((((S_3 \times C_3) : S_3) : S_3) : S_3) : S_3)$
T5.11	$((3^{1+2} : C_6) : S_3)$
T5.12	$((3^2 \times S_3) : S_3)$
T5.13	$(((3^{1+2} \times C_3) : S_3) : S_3)$
T5.14	$((((S_3 \times C_3) : S_3) : S_3) : S_3)$
T5.15	$(((C_4 \,.\, (2^4 : A_5)) : C_2) \times C_2)$
T5.19	$((Q_8 \,.\, \mathrm{PSL}(2,7)) : S_3)$
T5.21	$(((3^{2+6} : C_7) : C_3) : C_2)$
T6.1	$((2^{1+4} \times C_2) : S_3)$
T6.7	$((2^4 : S_3) : S_3)$
T6.9	$(((2^6 : C_7) : C_3) : C_2)$
T7.2	$((((2^4 : A_5) : C_2) : C_2) : S_3)$
T7.5	$((2^9 \,.\, \mathrm{PSL}(2,8)) : C_3)$
T7.7	$(((2^{1+4} : C_3) : S_3) : S_3)$
T8.4	$((((2^{1+6} : S_3) : S_3) : S_3) : S_3)$
T8.6	$(((2^{1+6} : C_3) : S_3) : S_3)$

Row	Certified shape
T8.7	$((((2^{1+6} : S_3) : S_3) : S_3) : S_3)$
T8.8	$(((C_2 \,.\, (A_5 \times (A_5 \times A_5))) : C_2) : S_3)$

5 Source and attribution boundary

The source is M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, the exact retained 21-page 2018 edition. The extension descriptions themselves are source assertions. The new work supplies their omitted explicit witnesses using the earlier saved images; it does not claim their historical discovery. The two previously proved order discrepancies, the local-period qualifications and all other source dispositions remain unchanged. The bounded prior review accepts only the exact presentations, geometry and previously verified named groups. It is not retroactively presented as acceptance of the structures checked here. There is no new Lean theorem, question solution or original credit in this supplement.

Correlation reductions and the exact square-torus parameters

Human Compression project

Abstract

We supply the definite diagram-(4) reduction stated on physical page 9 of Fernandes–Leemans–Weiss, and the definite presentation statements accompanying the experiments concerning diagram (5). Explicit inverse presentation maps give the parameter transformations. For every existing regular fork hypertope, its correlation extension has the intersection property and is the stated universal polytope. The group identities do not assert existence of a hypertope for every period or degenerate quotient.

1 The literal groups and their correlation

Write $a = \rho_0$, $b = \rho_1$, $c = \rho_2$, $d = \rho_3$, with b the central node of the fork. Its Coxeter relations are

$$a^2 = b^2 = c^2 = d^2 = (ab)^4 = (bc)^4 = (bd)^3 = (ac)^2 = (ad)^2 = (cd)^2 = 1. \tag{1}$$

Let H_s^0 be this presented group with $(abcb)^s = 1$, and let H_s^1 have instead $(abc)^{2s} = 1$, where s is a positive integer. These are the source's axial $(s, 0)$ and diagonal (s, s) square-torus residue relations, respectively. The assertion here concerns these entire presentations, not a finite image of them.

The substitution $\alpha : a \leftrightarrow c$, fixing b, d, preserves (1). Put $R = abcb$ and $Q = abc$. Then

$$\alpha(R) = cbab = bR^{-1}b, \qquad \alpha(Q) = cba = Q^{-1}.$$

It therefore preserves each of the two additional normal closures. It induces an automorphism of H_s^ϵ with square one, whether or not that presented group already defines a nondegenerate geometry. Form the external semidirect product

$$E_s^\epsilon = H_s^\epsilon \rtimes_\alpha \langle t \rangle, \qquad t^2 = 1.$$

Thus $tat = c$, $tbt = b$, $tdt = d$. The external factor has order two even in a collapsed small-parameter group. All statements about actual hypertope types below impose their stated nondegeneracy hypotheses.

Theorem 1 (Exact universal presentation reduction)**.** *On the ordered generators (t, a, b, d), E_s^0 is the Coxeter group $[4, 4, 3]$ with the extra relation $(tab)^{2s} = 1$, and E_s^1 is $[4, 4, 3]$ with the extra relation $(taba)^{2s} = 1$. In particular their marked local square-torus presentations are respectively (s, s) and $(2s, 0)$. In both cases H_s^ϵ is exactly the kernel of the character which sends t to -1 and a, b, d to 1.*

Proof. Eliminate c using $c = tat$. The relation $(ac)^2 = 1$ becomes $(at)^4 = 1$. The relations involving bc and cd are conjugates by t of those involving ba and ad; t commutes with b, d. The remaining relations are exactly the string $[4, 4, 3]$ relations on (t, a, b, d). For the periods the following are word identities, not order tests:

$$abcb = abtatb = (abt)^2 = t(tab)^2t, \tag{2}$$

$$(abc)^2 = abacbc = (abat)^2 = (taba)^{-2}. \tag{3}$$

In the first line commuting t and b gives $(abt)^2$. In the second line commute a, c, then use $tat = c$ and $tbt = b$. Taking the indicated powers proves equivalence of the extra relators and their full normal closures.

Conversely start with either stated string presentation, set $c = tat$, and retain t. The preceding steps give every fork relation, the correct period, and the conjugation rules for α. The resulting homomorphisms between that presentation and E_s^ϵ are inverse on all generators. This proves isomorphism of the presented groups, including the universal upper direction. Every defining string relator has even t-parity, so the character exists and is onto. In the semidirect product its kernel is precisely H_s^ϵ. Equivalently, the Schreier transversal $\{1, t\}$ gives generators a, c, b, d and the already displayed fork presentation; no omitted conjugate relator is required. □

The usual square-torus convention for a string marking (x, y, z) is $(xyzy)^q = 1$ for $(q, 0)$ and $(xyz)^{2q} = 1$ for (q, q). Thus (2) and (3), with $(x, y, z) = (t, a, b)$, give exactly the two parameter transformations in the theorem. This uses the same torus convention as the source and the retained square theory; it does not identify a torus by its group order. The convention can also be checked on the square plane: take

$$a(x, y) = (-x, y), \quad c(x, y) = (x, -y), \quad b(x, y) = (1 - y, 1 - x), \quad t(x, y) = (y, x).$$

Here $abcb$ translates by $(-2, 0)$, $(abc)^2$ by $(-2, 2)$, and $taba$ by $(1, -1)$. Rotation-conjugate period vectors give the complete two-dimensional period lattices. The string grid has basis $(1, -1), (1, 1)$, so its diagonal (s, s) lattice is the fork's axial $2s\mathbb{Z}^2$, while its axial $(2s, 0)$ lattice is generated by $(2s, -2s), (2s, 2s)$. This confirms both the scaling and the exchange of axial and diagonal labels. These coordinates are only the local square-plane model, not a purported model of the rank-four group.

2 Intersection property and the geometric assertion

A set of involutions has the intersection property if its generated subgroups satisfy $H_I \cap H_J = H_{I \cap J}$ for all subsets of its marking. The following elementary form of the usual parabolic patching argument suffices; no geometry is inferred just from an abstract group isomorphism. We retain the standard equivalence between string C-groups and regular abstract polytopes, and the regular-hypertope maximal-parabolic coset representation, as explicit foundational inputs.

Lemma 1 (Parabolic patching)**.** *If every maximal proper generated parabolic has the intersection property, and every two distinct such parabolics intersect in their common generated subgroup, then the entire marking has the intersection property.*

Proof. Let G_I, G_J be proper generated subgroups. Choose a missing marker i of I and a missing marker j of J. If $i = j$, the intersection property in that maximal parabolic applies. Otherwise an element of their intersection lies in the common parabolic G_T, where T omits i, j. Applying the intersection property first inside $G_{\bar{i}}$ and then inside $G_{\bar{j}}$ puts it in $G_{I \cap T}$ and $G_{J \cap T}$. Both are in G_T, which inherits the intersection property from a maximal parabolic. Their intersection is $G_{I \cap J}$, as required. Cases with I or J the full marking are immediate. □

Theorem 2 (From each existing fork to its polytope)**.** *Suppose $(H_s^\epsilon; a, b, c, d)$ is the actual marked group of a regular hypertope of source diagram (4), with the prescribed square-torus residue and the prescribed cubical residues. Then $(E_s^\epsilon; t, a, b, d)$ is a string C-group. Its polytope is respectively*

$$\big\{\{4, 4\}_{(s,s)}, \{4, 3\}\big\}, \qquad \big\{\{4, 4\}_{(2s,0)}, \{4, 3\}\big\},$$

and is universal with those marked local defining presentations. The given fork hypertope is recovered from its parity kernel and marking.

Proof. The marked group of the given regular hypertope has the intersection property. In $E = E_s^\epsilon$ put $L = \langle t, a, b\rangle$ and $K = \langle a, b, d\rangle$. The semidirect normal form gives

$$L \cap H_s^\epsilon = \langle a, b, c\rangle, \qquad \langle t, a\rangle \cap H_s^\epsilon = \langle a, c\rangle.$$

Indeed the listed kernel generators and t generate the respective subgroups, and t normalizes each listed kernel. Consequently

$$\langle t, a\rangle \cap \langle a, b\rangle = \langle a, c\rangle \cap \langle a, b\rangle = \langle a\rangle.$$

The two involutions a, c are independent commuting markers, so ta has order four. The remaining maximal subgroup of (t, a, b) is $\langle t, b\rangle = \langle t\rangle \times \langle b\rangle$. Its intersections with $\langle t, a\rangle$ and $\langle a, b\rangle$ are respectively $\langle t\rangle$ and $\langle b\rangle$, using parity and $\langle a, c\rangle \cap \langle b\rangle = 1$. Each two-marker subgroup has the intersection property, so Lemma 1 makes (t, a, b) a C-group. The other end deletion (a, b, d) inherits the intersection property and its string diagram directly from H_s^ϵ. Finally

$$L \cap K = \langle a, b, c\rangle \cap \langle a, b, d\rangle = \langle a, b\rangle.$$

For completeness the other two maximal parabolics are $M_1 = \langle t, b, d\rangle = \langle t\rangle \times \langle b, d\rangle$ and $M_2 = \langle t, a, d\rangle = \langle t, a\rangle \times \langle d\rangle$. The first product is direct by parity; the second uses $\langle a, c\rangle \cap \langle d\rangle = 1$ and the displayed commutations. Thus they too have the intersection property. In the ordering $(M_0, M_1, M_2, M_3) = (K, M_1, M_2, L)$ the six intersections are

(i, j)	$(0,1)$	$(0,2)$	$(0,3)$	$(1,2)$	$(1,3)$	$(2,3)$
$M_i \cap M_j$	$\langle b, d\rangle$	$\langle a, d\rangle$	$\langle a, b\rangle$	$\langle t, d\rangle$	$\langle t, b\rangle$	$\langle t, a\rangle$.

To verify the last three, both groups contain t, so intersect their parity kernels $\langle b, d\rangle$, $\langle a, c, d\rangle$, $\langle a, b, c\rangle$ inside H and then reattach $\langle t\rangle$. The first three follow in the same way with $K \subset H$. The original intersection property supplies every kernel intersection. Lemma 1 proves the full string intersection property. Its cubical vertex figure is the given (a, b, d) residue. Its facet group is the correlation extension of the given entire (a, b, c) torus group. Applying the rank-three presentation calculation of Theorem 1 identifies the facet with the stated torus, without an additional local quotient or period collapse.

Theorem 1 supplies exactly the global string relations and the defining facet period, with no further relators. Hence every regular polytope of those marked local types is a quotient, and the displayed polytope is universal. Conversely its exact parity kernel is the original marked group. A regular hypertope with that marking is its standard maximal-parabolic coset geometry, so this recovers the original geometry, not just an unmarked group order. □

The type exchange $a \leftrightarrow c$ is an actual correlation of the given regular fork geometry: on its maximal-parabolic cosets apply α and exchange the corresponding two types. This preserves coset intersection and is an involution. The new rank-four generating list is t, a, b, d. The source's three-letter parenthesis omits ρ_3; t, a, b generates the facet extension, while the complete rank-four assertion requires $d = \rho_3$ as well.

3 Diagram (5): definite algebra, separate geometric experiment

For diagram (5), keep the labels a, b, c of its square-torus path, with b central, and put d at the other vertex of the diamond. Its Coxeter relations are

$$a^2 = b^2 = c^2 = d^2 = (ab)^4 = (bc)^4 = (ad)^3 = (cd)^3 = (ac)^2 = (bd)^2 = 1. \tag{4}$$

This agrees with the actual source drawing: the two edges at b have label four and the two at d have the undisplayed label three. Let D_s^0, D_s^1 add, respectively, $R^s = 1$ and $Q^{2s} = 1$ to (4). The same substitution $a \leftrightarrow c$, fixing b, d, is an involutory automorphism of both groups.

Theorem 3 (Diamond-to-fork presentations)**.** *The extension $D_s^0 \rtimes \langle t \rangle$, marked (t, a, b, d) with a central, has exactly the diagram-(4) fork presentation with torus parameter (s, s). The extension $D_s^1 \rtimes \langle t \rangle$ has exactly that fork presentation with torus parameter $(2s, 0)$. In both cases D_s^ϵ is its exact t-parity kernel. These statements hold for all positive integers s as isomorphisms of the displayed presented groups.*

Proof. Again eliminate $c = tat$. The relation $(ac)^2$ gives $(ta)^4$; $(cd)^3$ is the t-conjugate of $(ad)^3$; $(bc)^4$ is the conjugate of $(ba)^4$. The remaining Coxeter relations give the fork with centre a, arms t, b of label four and arm d of label three. The period words transform by exactly (2)–(3). Thus they become $(tab)^{2s}$ and $(taba)^{2s}$, respectively. Now t, a, b is the fork's toroidal path, so these are its diagonal (s, s) and axial $(2s, 0)$ periods. The inverse substitution gives all original relations and the conjugation rules, proving both directions and the exact kernel assertion. In particular the source's conjugated word is correct: $\alpha(R) = cbab = (cbt)^2$ and $cbt = tab$. Conjugating $R^s = 1$ therefore gives precisely $(cbt)^{2s} = 1$. □

If the original diamond marking is a C-group, the extended fork marking also has the intersection property. The six maximal-parabolic intersections in the proof of Theorem 2 are unchanged. The one modification is $M_2 = \langle t, a, d \rangle$: it is now a rank-three string marking of type $\{4, 3\}$, rather than a direct product. Its three two-marker intersections follow from $\langle t, a \rangle \cap H = \langle a, c \rangle$, the original intersection property and $[t, d] = 1$, exactly as for $\langle t, a, b \rangle$. Lemma 1 applies first there and then to the full marking. This still does not prove flag transitivity of the new nonstring coset incidence system. The source explicitly calls hypertope production through this correlation experimental; the definite group and period assertions above do not remove that geometric qualification.

4 Exact scope and converse limitations

The group isomorphisms hold for every positive integer s. The geometric determination is quantified over each existing regular universal fork hypertope with its stated actual residues. It determines its marked group and coset geometry from the corresponding universal polytope. It does not claim that every period in the displayed presentations gives a hypertope, or that every quotient preserves its local types, intersection property or flag transitivity. In a proposed converse construction those properties must be established, not deduced merely from the parity map. For example, the separately proved bipartite halving theorem applies to a regular lattice polytope with a bipartite vertex graph; its actual lattice and incidence hypotheses must be checked before using it. No such unchecked hypothesis is inserted into Theorem 2.

This repairs the earlier combined documentary disposition for diagrams (4) and (5). Diagram (4) has the definite universal reduction just proved. Diagram (5)'s exact stated presentation and parameter transformations are proved separately in Theorem 3; its experimental hypertope-existence wording remains explicitly qualified. Neither an unproved flag-transitivity claim nor the order-768 future-work example is silently added to this supplement.

The source is M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, the exact retained 21-page 2018 edition, physical pages 9–10. The correlation idea and parameter reduction are theirs. The work here writes the inverse presentation maps and the missing marked-intersection argument, reusing the established Schreier and string-C-group principles. It claims no new original admission, historical priority, question solution or Lean formalization. No presentation enumeration or previously accepted geometric census is repeated.

A locally toroidal polytope with a nonorientable vertex figure

Human Compression project

Fernandes, Leemans and Ivić Weiss [1], pp. 18–19, mention the polytope $\{\{4,4\}_{(4,0)},\{4,6\}_3\}$ and state that its group has order 768 and its vertex figure has nonorientable genus four. They explicitly leave further study of this example aside. We verify these assertions separately from their tables of orientably regular and chiral examples. Here the subscript in $\{4,6\}_3$ is the Petrie length.

Proposition 1. *Let U be the group generated by involutions a,b,c,d with*

$$(ab)^4=(bc)^4=(cd)^6=(ac)^2=(ad)^2=(bd)^2=1,\qquad (abcb)^4=(bcd)^3=1.$$

Then U, with this ordered generating set, is a string C-group of order 768. Its facets are $\{4,4\}_{(4,0)}$, and its vertex figures are the nonorientable maps $\{4,6\}_3$ of genus four. It is universal for these two prescribed local presentations.

Proof. The program `verify.g` forms the displayed finitely presented group itself. GAP's exact finite group algorithms give $|U|=768$ and a bijective homomorphism to a permutation group of degree 16. For clarity, the images of the four distinguished generators are

$$\begin{aligned}
a&=(1\,2)(3\,6)(4\,7)(5\,8)(9\,11)(10\,12)(13\,14)(15\,16),\\
b&=(1\,3)(2\,6)(4\,5)(7\,8)(9\,13)(10\,14)(11\,15)(12\,16),\\
c&=(1\,4)(2\,7)(3\,9)(5\,10)(6\,11)(8\,12)(13\,14)(15\,16),\\
d&=(1\,5)(2\,8)(3\,4)(6\,7)(9\,12)(10\,11)(13\,16)(14\,15).
\end{aligned}$$

This permutation image alone would give only a lower bound on the universal group. The calculation of the finitely presented group supplies the upper bound and checks bijectivity. For every pair of subsets $I,J\subseteq\{a,b,c,d\}$, the program checks

$$\langle I\rangle\cap\langle J\rangle=\langle I\cap J\rangle.$$

These are all 256 pairs, including the empty and full subsets. The adjacent product orders are exactly $4,4,6$. The standard coset construction for string C-groups therefore gives a regular abstract polytope with this group and marking.

The facet and vertex stabilizers $\langle a,b,c\rangle$ and $\langle b,c,d\rangle$ have orders 128 and 48. Independently forming their three-generator local presentations gives these same orders; the maps sending local generators to the displayed distinguished generators are checked bijective. Thus there are no additional local relations. The first presentation is the square torus $\{4,4\}_{(4,0)}$: in the affine square reflection group, *abcb* is a unit translation, and its fourth power and its conjugates generate the period lattice $4\mathbb{Z}^2$. The quotient has order $8\cdot16=128$. The second is exactly the Petrie quotient $\{4,6\}_3$; the product *bcd* has order three.

In the vertex figure the numbers of vertices, edges and faces are respectively

$$[\langle b,c,d\rangle:\langle c,d\rangle]=4,\qquad [\langle b,c,d\rangle:\langle b,d\rangle]=12,\qquad [\langle b,c,d\rangle:\langle b,c\rangle]=6.$$

The Euler characteristic is therefore $4-12+6=-2$. The relation $(bcd)^3=1$ is a closed walk of odd length nine in its flag graph. Orientability would assign opposite orientations to adjacent

flag triangles and hence make this graph bipartite. The odd closed walk rules this out. The map is a connected closed nonorientable surface, so its genus is $2 - (-2) = 4$.

Finally, any regular polytope with these local presentations satisfies the displayed defining relations. Its distinguished generators therefore receive a surjective homomorphism from U. Since U itself has the required intersection property and local groups, it is the universal object for these local presentations. □

The group order and genus are assertions of the cited source, not new claims of discovery. The new work is their explicit verification in the present proof system. The finite presentation, all subgroup intersections and the local isomorphisms are required exact computations in GAP; this note is not a complete Lean proof. It settles this isolated source assertion and does not answer any of that paper's open questions.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, the retained 21-page 2018 edition, pp. 18–19.